

Active Management of IO Beta

Cello Research

March 1, 2016

IO Beta Allocation Strategy: February 2016

Commentary

- ▶ **IO Index Returns:** -1.39% for IO Index with TBA hedges, 0.52% for Index with Curve and TBA Hedges (February 2016)
- ▶ **Fund Beta Allocation YTD:** Outperformed IO Index (w/ Curve Hedges) by 138bps

IO Beta Allocation Strategy: March 2016

Summary

- ▶ **Actual Allocation:** 32%
- ▶ **Model 1 (IO Px/Cpn not duration-adjusted):** 36%.
(February: 17%)
- ▶ **Model 2 (IO Px/Cpn duration-adjusted):** 33%.
(February: 53%)

Active Management of IO Beta

Investigate if we can design an **active** strategy for investing in IO Beta that outperforms a **passive** strategy that is always 100% invested.

Metrics and Set-up

- ▶ Create a **Hedged IO** index
 - ▶ Hedge various IOS with current coupon TBA using model durations (Barclays)¹
 - ▶ Index is created by using monthly market value weights for each IOS based on the market weight of the corresponding cohort
 - ▶ Ginnie Mae IOS are excluded
- ▶ Simulating Passive IO Beta: Total Return and Sharpe Ratio of Hedged IO index from March 2011 - February 2016
- ▶ Simulating Active IO Beta: Dynamic allocation to Hedged IO index based on relative value metrics

¹Note that we do *not* hedge out the curve risk and volatility risk associated with an individual IOS.

Performance of Passive IO Beta Strategy: Mar 2011 - Feb 2016

- ▶ **CAGR:** 8.2%
- ▶ **Ann. Volatility:** 16.5%
- ▶ **Sharpe Ratio:** 0.6
- ▶ **% Positive Months:** 58
- ▶ **Best Month:** 11.6%
- ▶ **Worst Month:** -12.9%
- ▶ **Max Drawdown:** -23.5%
- ▶ **Drawdown Period:** June 2011 - March 2013

Framework for Active IO Beta Strategy

- ▶ Regress Hedged IO Index returns against RV metrics (1m lag):
 - ▶ IO Index Price/Coupon²
 - ▶ Projected 1-month Carry³
 - ▶ Convexity: 60-day moving average of realized volatility on the current coupon
 - ▶ Prepayment Risk: Incentive for the Mortgage Universe
 - ▶ Yield Curve Slope: 10s-2s slope
- ▶ Calculate z-score of projected return and allocate to Beta (Net IO) based on score

²While OAS would also be useful, in practice dealer model updates are not always appropriately back-filled which makes it challenging to get a consistent time series of values.

³This is not available on a historical basis so we use realized carry as a proxy.

Active IO Beta Strategy: Regression Results

- ▶ Significant Variables
 - ▶ High: IO Index Price/Coupon, Prepayment Risk
 - ▶ Medium: Realized Volatility
 - ▶ Low: Realized Carry, Yield Curve Slope, Index OAS
- ▶ Final Model
 - ▶ Proj. Return $\sim P_x/Cpn + Cnvx + PrepayRisk + Carry$

Active IO Beta Strategy: Beta Allocation

- ▶ Solve for $k = a + bz$ ($z = z\text{-score}$) such that we optimize the Sharpe Ratio of the strategy $k \times \text{Hedged IO Return}$.
- ▶ Set $\text{IO Beta}_{\max} = 100\%$ and $\text{IO Beta}_{\min} = -0.25\%$
- ▶ Rescale k so that it is (more or less) mapped to these boundaries.

Active IO Beta Strategy: Two Beta Allocation Models

- ▶ Model 1: Don't adjust IO Index P_X/C_{pn} for Interest Rates. $R^2 = 0.26$.
- ▶ Model 2: Adjust IO Index P_X/C_{pn} for Interest Rates.⁴ $R^2 = 0.17$.

⁴We estimate that the P_X/C_{pn} ratio moves by about 0.14 for each 10bps move in the current coupon.

Active IO Beta Strategy: Model 1 Return Profile

- ▶ **CAGR:** 8.9%
- ▶ **Ann. Volatility:** 5.8%
- ▶ **Sharpe Ratio:** 1.5
- ▶ **% Positive Months:** 68
- ▶ **Best Month:** 9.5
- ▶ **Worst Month:** -1.1%
- ▶ **Max Drawdown:** -2.0%
- ▶ **Drawdown Period:** November 2011 - March 2012

Active IO Beta Strategy: Model 2 Return Profile

- ▶ **CAGR:** 7.1%
- ▶ **Ann. Volatility:** 6.1%
- ▶ **Sharpe Ratio:** 1.2
- ▶ **% Positive Months:** 71
- ▶ **Best Month:** 9.4
- ▶ **Worst Month:** -3.0%
- ▶ **Max Drawdown:** -3.6%
- ▶ **Drawdown Period:** November 2011 - March 2012

Track Record of IO Beta Allocation Strategies

	IO Beta Allocation			Monthly Hedged Return					Yearly Return				
	Fund	Model1	Model2	100% Beta	100% Beta CrvAdj	Fund	Model1	Model2	100% Beta	100% Beta CrvAdj	Fund	Model1	Model2
Jan-15	26%	-11%	-16%	-8.8%	-8.9%	-2.31%	0.98%	1.40%					
Feb-15	25%	26%	59%	8.2%	5.5%	1.38%	1.44%	3.24%					
Mar-15	21%	-7%	-51%	-0.4%	1.2%	0.25%	-0.08%	-0.61%					
Apr-15	15%	-21%	-16%	4.5%	1.2%	0.18%	-0.26%	-0.19%					
May-15	22%	-28%	-36%	0.1%	-1.7%	-0.37%	0.47%	0.61%					
Jun-15	31%	-17%	-24%	1.6%	-0.7%	-0.20%	0.11%	0.16%					
Jul-15	33%	-25%	-51%	-0.1%	1.2%	0.40%	-0.30%	-0.61%					
Aug-15	28%	-42%	-16%	-0.3%	-1.5%	-0.42%	0.64%	0.23%					
Sep-15	24%	-24%	-23%	-0.5%	-1.6%	-0.38%	0.39%	0.37%					
Oct-15	26%	-10%	5%	-0.3%	-0.1%	-0.03%	0.01%	-0.01%					
Nov-15	21%	2%	-5%	1.5%	1.8%	0.38%	0.04%	-0.08%					
Dec-15	22%	15%	0%	0.1%	1.4%	0.30%	0.21%	0.01%	5.61%	-2.23%	-0.85%	3.66%	4.53%
Jan-16	17%	12%	3%	-4.6%	-2.1%	-0.36%	-0.25%	-0.07%	-5.99%	-1.58%	-0.20%	-0.16%	0.21%
Feb-16	31%	17%	54%	-1.4%	0.5%	0.16%	0.09%	0.28%					
Mar-16	32%	36%	33%										

As of: Mar 1, 2016

Evolution of IO Beta Allocation RV Variables

Date	MthlyHedgedReturn	OAS	Carry	RealizedVol	YCSlope	PrepayRiskPrem	PxCpnMult	UB1m	RefIPctPool	RefIPctLoan	RefIndex	CC	CCChanges	CurveHedgedReturn
1/30/2015	-11.82	65	1.10	60	121	103	4.78	0.17	0.66	0.66	2585	2.45	-38	-8.94
2/27/2015	11.16	-87	1.62	73	138	89	5.51	0.17	0.61	0.56	1740	2.74	28	5.53
3/31/2015	-0.70	-97	0.99	78	138	92	5.37	0.18	0.64	0.62	1942	2.66	-8	1.18
4/30/2015	5.21	28	0.24	69	147	81	5.56	0.18	0.64	0.61	1559	2.75	9	1.19
5/29/2015	1.25	27	0.37	68	149	73	5.66	0.18	0.55	0.50	1234	2.81	6	-1.66
6/30/2015	4.68	10	0.32	75	170	51	6.16	0.19	0.45	0.40	1246	3.07	26	-0.66
7/31/2015	-1.55	14	0.19	82	153	66	5.93	0.19	0.50	0.42	1418	2.92	-16	1.18
8/31/2015	0.14	21	0.49	76	146	68	5.92	0.20	0.55	0.53	1546	2.93	2	-1.53
9/30/2015	-1.25	40	0.61	67	142	71	5.70	0.19	0.52	0.49	1590	2.8	-13	-1.59
10/30/2015	0.19	42	0.56	66	142	68	5.69	0.19	0.58	0.55	1535	2.84	4	-0.13
11/30/2015	2.39	3	0.55	58	129	57	5.89	0.24	0.48	0.45	1489	2.95	10	1.78
12/31/2015	1.14	5	0.59	61	122	53	5.97	0.43	0.45	0.41	1667	3.01	6	1.36
1/29/2016	-4.64	21	0.74	57	116	74	5.38	0.43	0.62	0.60	1734	2.65	-36	-2.11
2/29/2016	-1.39	-7	1.58	67	95	92	5.07	0.44	0.67	0.63	2100	2.53	-12	0.52

Allocation Grid for IO Basis

	PxCpnMult					
PrepayRiskPrem	5.13	5.23	5.33	5.43	5.53	5.63
72.7	68	58	48	39	29	19
77.7	58	48	38	28	18	8
82.7	47	37	28	18	8	-2
87.7	37	27	17	7	-3	-13
92.7	26	17	7	-3	-13	-23
97.7	16	6	-4	-14	-24	-34